

SAMARA SMYSHLYAEVKA, Russian Federation

WMO: 288070

Lat: 53.250N

Lon: 50.450E

Elev: 40

StdP: 100.85

Time Zone: 4.00 (E04)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.5	-25.3	-31.5	0.2	-28.4	-28.2	0.3	-25.1	10.5	-6.1	10.1	-6.7	1.2	170	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	33.4	20.7	31.3	20.4	29.5	19.6	22.1	30.0	21.3	29.3	20.5	27.9	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	25.5	18.6	13.5	24.4	17.7	12.8	23.7	64.6	30.0	61.8	29.2	58.9	27.9	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.0	7.8	-31.3	35.9	2.8	1.8	-33.4	37.1	-35.0	38.2	-36.6	39.2	-38.7	40.5		
(5)			-31.5	23.9	2.7	1.2	-33.4	24.7	-35.0	25.4	-36.5	26.1	-38.5	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-10.5	-10.9	-3.9	6.9	14.1	18.8	21.6	19.4	13.0	6.1	-1.8	-8.5		
	DBStd	12.84	7.45	7.00	5.21	5.59	4.64	4.33	3.57	4.07	3.96	4.31	6.10	7.80		
	HDD10.0	2871	635	585	429	124	17	1	0	14	136	356	573			
	HDD18.3	4965	893	818	688	344	144	46	12	37	167	379	605	832		
	CDD10.0	1207	0	0	0	31	143	266	358	291	103	15	0	0		
	CDD18.3	261	0	0	0	1	13	60	112	69	6	0	0	0		
	CDH23.3	3059	0	0	0	19	211	674	1257	797	101	0	0	0		
	CDH26.7	1172	0	0	0	2	51	245	519	336	19	0	0	0		
(14) Wind	WSAvg	3.2	3.2	3.3	3.6	3.6	3.7	3.1	2.7	2.8	2.7	3.2	3.6	3.3		
(15) Precipitation	PrecAvg	530	45	34	31	37	29	58	59	47	42	52	48	48		
	PrecMax	648	91	91	87	98	67	168	118	108	104	106	151	133		
	PrecMin	285	14	0	3	1	1	8	0	0	12	8	9			
	PrecStd	77	20	22	19	26	19	37	35	30	28	26	30	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	2.4	11.7	25.7	30.5	35.2	36.3	36.3	29.0	21.1	11.7	4.5		
		MCWB	1.6	1.1	7.3	14.4	17.6	20.1	20.6	20.6	16.3	12.6	10.2	3.5		
	2%	DB	1.7	1.7	6.9	22.5	28.0	32.2	33.3	32.9	26.4	18.2	8.7	2.6		
		MCWB	0.8	0.7	3.7	12.8	16.9	20.0	21.2	20.2	16.1	11.3	7.0	1.5		
	5%	DB	0.9	1.0	4.1	19.8	25.7	29.4	31.4	30.4	23.5	15.3	6.8	1.7		
		MCWB	0.3	0.3	2.0	11.6	15.6	19.6	20.9	19.9	15.4	10.3	5.6	0.8		
	10%	DB	-0.3	-0.4	2.6	16.7	23.3	27.1	29.7	27.8	20.7	13.1	5.2	1.0		
		MCWB	-0.9	-1.1	1.1	10.0	14.6	18.6	20.2	18.8	14.1	9.5	4.1	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	1.4	7.4	15.6	19.7	22.2	23.7	22.4	18.8	14.2	10.8	4.0		
		MCDB	2.6	2.0	11.8	23.7	27.5	30.9	31.0	31.6	24.8	17.3	11.7	4.4		
	2%	WB	1.0	0.9	3.8	13.9	17.9	21.2	22.6	21.4	17.2	12.4	7.4	1.8		
		MCDB	1.5	1.5	6.5	21.1	25.7	29.6	30.0	30.6	23.5	16.4	8.5	2.4		
	5%	WB	0.3	0.3	2.2	12.0	16.7	20.3	21.7	20.5	16.0	10.9	5.7	0.9		
		MCDB	0.9	1.0	3.6	18.6	24.4	27.6	29.0	29.1	22.0	14.5	6.7	1.5		
	10%	WB	-1.0	-1.2	1.2	10.3	15.4	19.4	21.0	19.4	14.7	9.7	4.2	0.3		
		MCDB	-0.3	-0.4	2.4	16.3	21.5	25.9	28.3	26.5	19.9	12.8	5.1	0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.3	8.4	10.7	12.4	12.0	12.3	12.6	11.8	8.8	5.3	6.0		
		MCDBR	5.5	5.3	8.5	16.1	16.4	15.2	15.3	16.6	16.9	13.9	6.4	4.2		
	5% WB	MCWBR	5.2	4.8	5.9	8.0	7.4	6.3	6.0	6.6	8.2	8.0	5.5	4.0		
		MCWBR	5.5	5.0	7.8	14.5	15.0	13.1	13.3	15.0	14.6	11.0	5.7	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.328	0.375	0.432	0.408	0.401	0.411	0.430	0.385	0.355	0.331	0.303		
		taud	2.063	2.031	1.995	2.086	2.271	2.343	2.312	2.240	2.338	2.332	2.235	2.085		
	Ebn at Noon		625	739	784	792	841	851	833	790	783	721	607	546		
		Edh at Noon	75	106	137	143	125	118	120	122	98	79	64	61		
(44) All-Sky Solar Radiation	RadAvg		0.92	1.99	3.34	4.34	5.64	6.08	6.01	4.92	3.25	1.64	0.70	0.51		
	RadStd		0.15	0.20	0.38	0.45	0.42	0.67	0.51	0.40	0.46	0.28	0.12	0.14		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	+1.09	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page